Stress-Testing USDFC:

Building a Robust FIL-Backed Stablecoin for the Future

The CEL team

Remark. *These notes are a simplified version of our more extensive stress-testing report presented here.*

In a world where cryptocurrency prices can swing wildly within minutes, stable and reliable digital assets provide an essential utility to ecosystems. For Filecoin, whose network underpins a data storage economy where providers and clients need predictable costs and revenues, this stability is particularly valuable. Here USDFC provides one answer—a stablecoin designed to maintain a one-to-one peg with the U.S. dollar while being backed by Filecoin (FIL) collateral. In these notes, we present the simulation methodology we developed to test USDFC's robustness under a variety of challenging market scenarios.

Demystifying USDFC

USDFC is a stablecoin that remains pegged to the US dollar by being fully collateralized with Filecoin (FIL). Its design is inspired by successful Ethereum protocols such as MakerDAO's DAI and Liquity's LUSD. In this system, users deposit FIL into a smart contract—called a trove—to mint USDFC. The key to safety in this model is the collateral ratio (CR), which compares the current FIL value to the amount of USDFC issued. If the CR drops below a preset threshold (say 110%), the trove is vulnerable to liquidation, meaning that the collateral is sold off to cover the debt.

Consider a simple example with Alice and Bob. Alice holds FIL and decides to open a trove. She deposits her FIL and mints USDFC, using her collateral to back the newly created tokens. For the protocol to work, Alice must maintain her collateral ratio above the minimum threshold. However, if the price of FIL drops unexpectedly, Alice's CR could fall, triggering a liquidation. In this event, the protocol automatically seizes a portion of her FIL to cover the USDFC that she borrowed, protecting the system's overall health.

Now, imagine that due to sudden market volatility, USDFC starts trading below $1. The system is designed to encourage arbitrage to restore the peg—a process known as repegging. Here, Bob enters the picture. As an arbitrageur, Bob buys USDFC at a discount from the market, then redeems it by returning the tokens to the protocol in exchange for the underlying FIL collateral. The redemption process typically targets troves with the lowest collateral ratios first. By redeeming USDFC and receiving FIL, Bob effectively helps push the stablecoin's price back to its intended $1 mark. This repegging mechanism is critical because it ensures that USDFC maintains stability even when market forces pull its price away from the peg.

However, the redemption process is not without risks. While redemptions serve as a self-correcting mechanism, they also come with potential downsides for trove owners like Alice. If forced redemption events occur too frequently, Alice's collateral could be rapidly reduced—sometimes in a cascade—leaving her exposed to losses. To mitigate these risks, the system can incorporate additional strategies. For example, users might establish a buffer trove: an intentionally weaker position with a slightly lower CR that absorbs redemptions in place of the primary trove. This approach helps protect a user's main collateral while still enabling the system to maintain the USDFC peg.

In essence, USDFC is built to handle market volatility by using automatic liquidations and redemption mechanisms. The repegging process, primarily driven by arbitrageurs like Bob, ensures that even when USDFC trades below $1, the price correction is swift and self-sustaining. At the same time, the system designers are mindful of redemption risks and have outlined methods to protect individual users from the negative consequences of forced sales. There has to be a careful balance of stability and risk management for USDFC to be a robust option for the Filecoin ecosystem.

From the previous discussion, we can summarize the main potential *risks* associated to a CDP:

1. **(Mis)Liquidation Risk**, which refers to the risk of being liquidated (as a user) or failing to liquidate (for the protocol) should the collateral drop below the given over collateralization ratio,
2. **Redemption Risk,** which occurs when someone else *pays* part of your debt, which can in turn be seen which forces you to realize any capital gains on your token.
3. **Depeg Risk,** which corresponds to the event where the price of USDFC goes significantly below (or above) $1

In order to properly investigate these risks, here at CEL we developed an independent simulation methodology – effectively a digital twin – of USDFC to stress test it on several scenarios and market conditions. We describe our simulation methodology below. Our findings indicate USDFC is robust under different scenarios, but also offer some recommendations on how to further minimize these risks.

Remark: *The simulation environment (together with instructions on how to use it) can be found here, and the reader is encouraged to test it independently.*

The Simulation Methodology: An Intuitive Approach

Our simulation environment serves as a "digital twin" of the USDFC system, allowing us to test its performance under a broad range of market conditions. At a high level, the simulator recreates what happens in the real world by mimicking the unpredictable nature of Filecoin

(FIL) prices and user behaviors. By doing so, we can estimate how the stablecoin responds to everyday market movements as well as extreme events.

The simulator relies on several core components. First, to model FIL's price dynamics, we use Geometric Brownian Motion (GBM). Imagine FIL's price as a ship navigating stormy seas. While the ship generally follows a predictable course (the drift), it is buffeted by random waves (volatility). GBM captures both this steady drift and the random shocks, thereby providing a realistic model of price paths over time. These simulated price paths form the backbone of the entire framework, influencing every subsequent operation.

Next, we simulate individual user positions—called troves—where FIL is deposited and USDFC is minted. For each trove, the simulator continuously computes the collateral ratio, which is the ratio of the current FIL value to the debt in USDFC. This ratio is critical because it determines whether a trove remains healthy or becomes subject to liquidation. If a trove's collateral ratio drops below a preset threshold, the simulator automatically triggers a liquidation event. The system then "seizes" part of the collateral to cover the USDFC debt, mimicking real-world risk management strategies.

Redemption is another essential process in our simulations. Whenever USDFC's market price slips below $1, the system activates a mechanism that allows traders to buy USDFC at a discount and redeem it for FIL. This process, known as repegging, helps restore the price back to its target level. In our simulation, we track these redemption events—identifying which troves get affected and how the overall supply adjusts—to ensure that the repegging mechanism is both effective and fair under various market scenarios.

The simulation methodology is implemented through thousands of Monte Carlo runs. Each simulation run uses different random inputs to replicate thousands of possible market paths. This approach provides us with statistical insights, such as an estimate of the probability of liquidation, frequency of forced redemptions, and overall stability of the USDFC peg (conditional on the assumptions of the model). These insights allow us to optimize key parameters, like the minimum collateral ratio, ensuring that the system remains resilient whether the market is bullish, bearish, or highly volatile.

Below, we plot some exemplary results of the market scenarios (*bullish, bearish, base and high-volatility of FIL price)* and observables that we can obtain with our simulation environment,

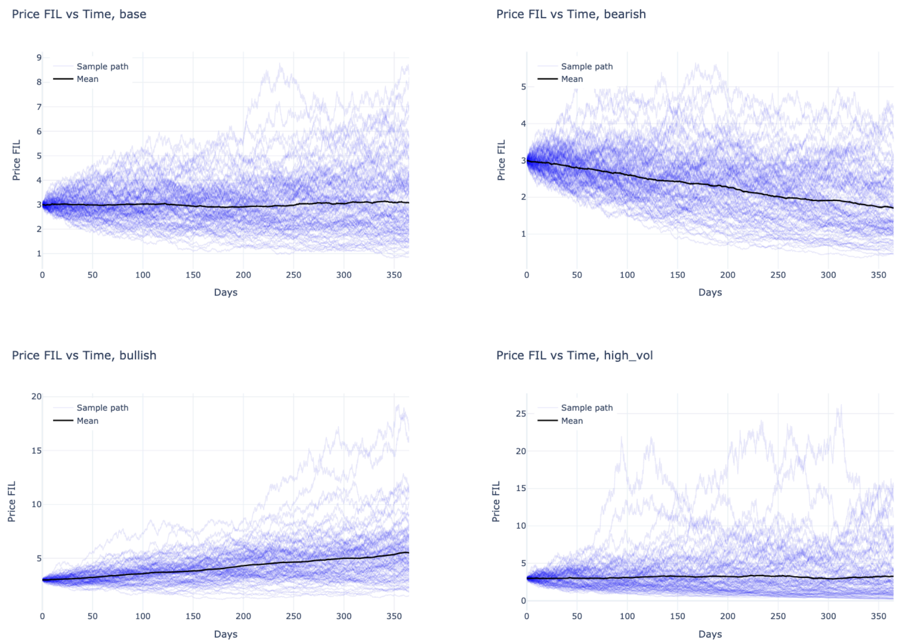

Figure 1: Different price-paths realisations for FIL under different market conditions; base (top left), bearish (top right), bullish (bottom left) and high-volatility (bottom right).

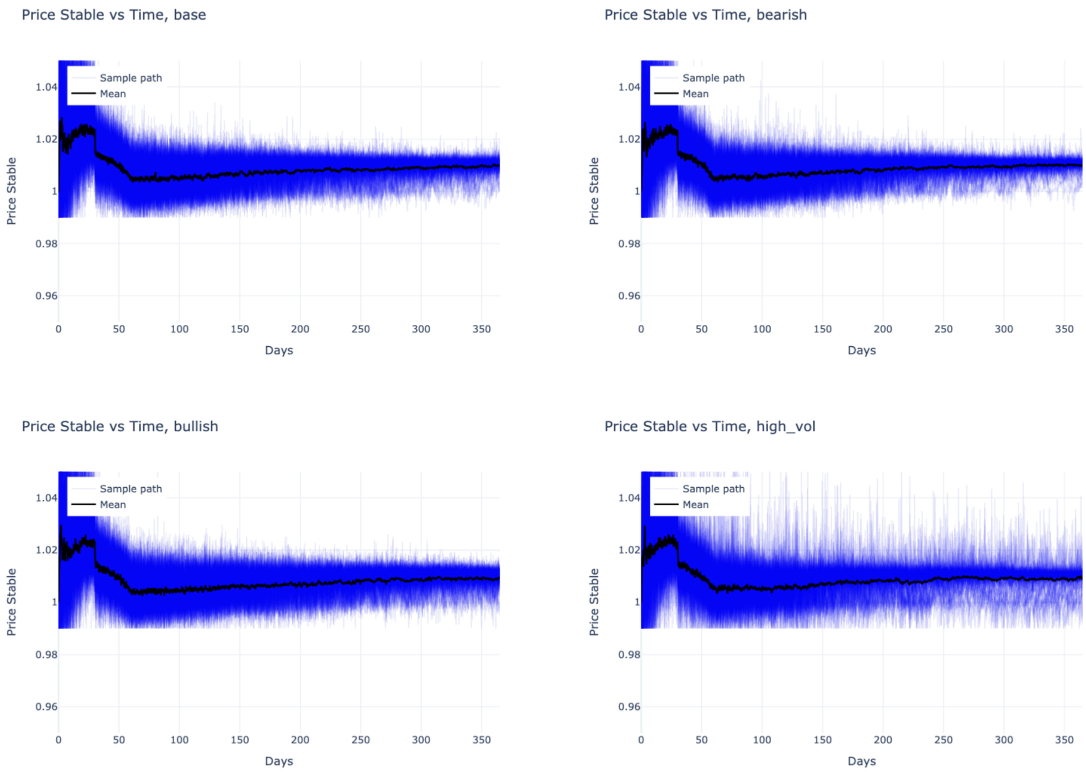

Figure 2: Different price-paths realisations for USDFC under different market conditions;

base (top left), bearish (top right), bullish (bottom left) and high-volatility (bottom right).

Notice that in Figure 2, we stress test the depeging risk and observe that, while different market conditions might increase the volatility of the USDFC token, it still remains pegged around $1 on an acceptable manner.

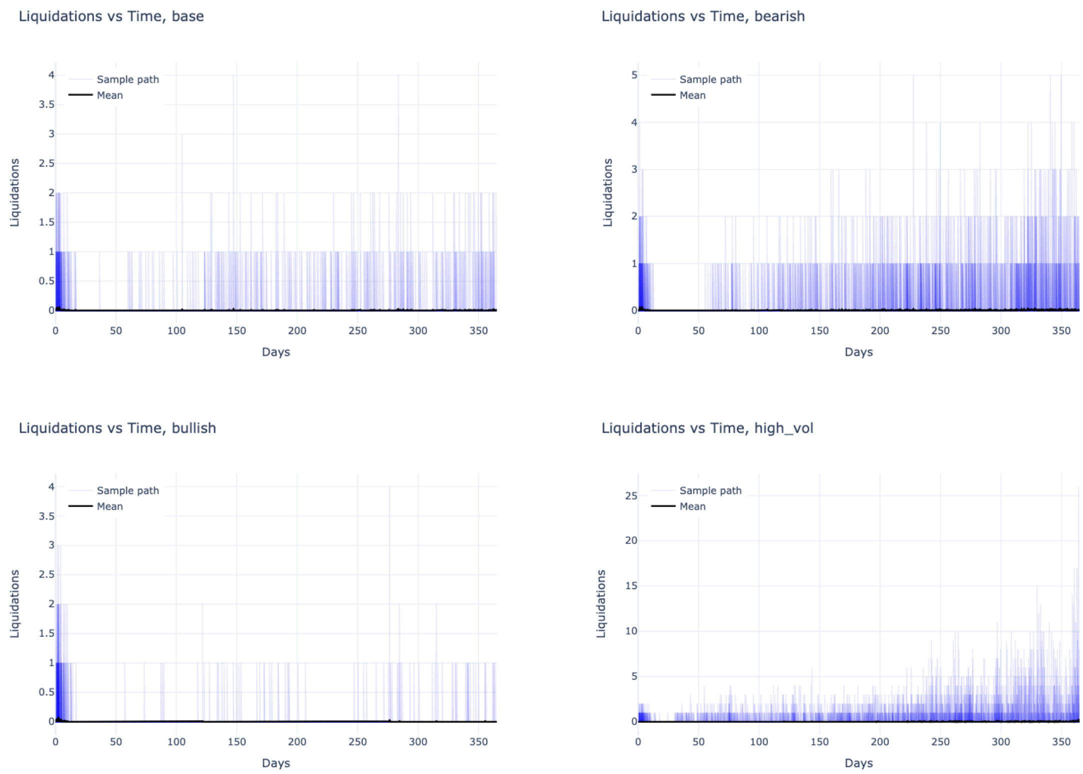

Figure 3: Liquidations vs Time under different market conditions; base (top left), bearish (top right), bullish (bottom left) and high-volatility (bottom right).

Robustness Under Stress

Our simulations reveal that the USDFC protocol is stable under a wide range of market conditions that are modeled. By subjecting the system to a range of scenarios—from moderate, everyday volatility to sudden and dramatic crashes of FIL's value—we have demonstrated that USDFC maintains its fundamental design integrity, keeping the stablecoin's peg and overall stability intact.

Imagine FIL's price experiencing a rapid and severe decline—say, a 90% drop. In such scenarios, our simulator shows that the system's built-in mechanisms kick in. In the model, the liquidation process, which protects the overall collateralization of the system, is triggered for troves that fall below the safe threshold. This early warning system prevents cascading failures where multiple troves could otherwise collapse simultaneously. In addition, if USDFC begins trading below $1, the repegging mechanism encourages third-party participation. Arbitrageurs step in by buying USDFC at a discount and redeeming it for FIL, which helps restore the price back to its intended peg.

Crucially, the stress tests indicate that the performance of USDFC improves markedly with prudent adjustments to protocol parameters. For example, while a collateral ratio of 110% might be sufficient under stable market conditions, increasing the ratio to around 150% during high volatility drastically reduces the frequency of forced liquidations. This additional margin acts as a buffer, allowing the system more leeway to absorb shocks without triggering disruptive events. These measures ensure that, even in the worst-case scenarios, individual users' risks are compartmentalized and do not undermine the overall stability of the protocol.

In summary, even when facing extreme market shocks, USDFC demonstrates a resilient capacity to maintain its peg through timely liquidations, effective repegging, and well-calibrated collateral protection, which would make it a reliable tool for ensuring stability in the fast-paced world of decentralized finance.

Historical Analysis and Liquidation Risk Assessment

We performed a two-step process to test the (mis)liquidation risk. On the one hand, we used our simulation environment to examine how the probability of being liquidated under different market scenarios, depended on the collateralization ratio. Our experimental results show that, while higher CRs result in lower liquidation probabilities, these are still rather low and, all else being equal, lower CRs are preferable as they lead to higher capital efficiency.

In addition to this, our study of historical Filecoin price data forms the backbone of our understanding of liquidation risk within the USDFC system. Over the course of one year—from April 2024 to April 2025—we collected high-frequency price data to capture the real-world volatility of FIL. This extensive dataset allowed us to measure price movements over various block intervals and understand how extreme fluctuations might trigger liquidations.

In our analysis, we divided the data into several block intervals—1, 2, 10, and 100 blocks. For each interval, we calculated key statistics such as the average percentage change, standard deviation, and the extreme minimum and maximum price changes. The intuition here is straightforward: the shorter the interval, the smaller the price change on average. However, when we extend the observation window (for example, to 100 blocks), the price variations become much more pronounced, revealing the potential for larger drops.

The results revealed that short-term price movements (1-10 block intervals) presented noticeable liquidation risk, with instances of price drops exceeding 10%. At the 1-block interval, price changes averaged -0.0055% with a standard deviation of 0.3488%, indicating relatively stable short-term price behavior but with a maximum negative movement of -22.8831%. Similarly, 2-block and 10-block intervals showed increasing volatility with maximum negative price movements of -22.9095% and -43.543% respectively.

Block Interval	Mean Change (%)	Std Dev (%)	Min (%)	Max (%)	Liquidation Risk (%)
1	-0.0055	0.3488	-22.8831	1.8047	0.019
2	0.0105	1.4375	-22.9095	98.2385	0.0381
10	0.0302	3.4635	-43.543	130.797	0.2283
100	0.2044	7.8568	-48.8413	150.253	1.1035

At the 100-block interval, we observed the highest liquidation risk in our dataset. The standard deviation increased to 7.9\%, with a minimum price change of -48.8\%, representing the largest downturn observed. This resulted in a liquidation risk of 1.1\%, indicating that approximately 1 in 91 positions would face liquidation at this interval with the current CR.

Based on the statistical properties of the observed price volatility, particularly the 100-block interval data, **we can conclude that a current 110% CR should be enough for the intended purposes, provided that there's sufficient *liveness* in the liquidation bots (i.e., that such a liquidation is triggered within 100 blocks after dropping from 100% CR). This can comfortably be managed by the protocol.**

Managing Redemption Risk

In the operation of USDFC, redemption is a critical process that serves to restore the stablecoin's peg when market forces push it below $1. However, the mechanism that enables redemptions—while essential—also introduces the risk that trove owners may face forced collateral sales at unfavorable times. If the market slides and redemptions are triggered, the protocol prioritizes liquidating positions with the lowest collateral ratios. This means that users with troves that are too close to the minimum threshold could see their collateral rapidly depleted, potentially resulting in significant losses.

To address this challenge, our research explores multiple strategies designed to mitigate the negative impact of forced redemptions. One effective approach is the use of buffer troves. In this setup, a user opens an additional trove with a deliberately lower collateral ratio than their primary position. When redemptions occur, the protocol naturally targets the buffer trove first because it represents the weakest link. This arrangement not only shields the user's main trove from immediate liquidation but also provides time for the user to take corrective action. By absorbing redemption pressure through a buffer trove, the overall risk to the user’s substantial collateral is reduced, preserving long-term stability and maintaining confidence in the system.

Overall, the simulation studies demonstrate that while redemption risk is inherent in any collateralized stablecoin design, it can be actively managed through practical measures. Whether it is employing buffer troves, or adjusting the fee and collateral parameters, these methods work in tandem to reduce redemption exposures and maintain the integrity of the peg. As we continue to stress-test and refine these strategies, our ultimate goal remains to protect users while ensuring that USDFC consistently performs its fundamental role as a stable, reliable, and secure digital asset in the Filecoin ecosystem.

The So What: Transparent Stress Testing

By subjecting USDFC to a battery of open-source simulations, we provide a transparent, neutral, and data-backed assessment of its robustness under stress. This independent methodology means that investors, developers, and the broader Filecoin community can check for themselves how USDFC was tested, and test their own possible scenarios to gain confidence.

Our open-source simulation environment, available on GitHub at github.com/juanpmcianci/SF, allows interested parties to replicate our tests, tweak parameters, and even run additional scenarios. We hope and invite other researchers and users to test for themselves their own assumptions and scenarios, to determine their level of confidence both in our methodology and USDFC.

Disclaimer: The simulations and analyses presented in this document are based on mathematical models that, while rigorous, are inherently approximations of real-world market behavior. These models make certain assumptions about price movements, market dynamics, and participant behavior that may not fully capture all possible scenarios. The information provided herein is for educational and research purposes only and should not be construed as financial, investment, or trading advice. Readers should conduct their own research!